

Question

A peptide compound with antitumor activity was isolated from a marine alga. The compound was subjected to sequence analysis using chemical methods, and the experiments and results are as follows:

1. After treating the compound with 2-mercaptoethanol, sequencing was performed using Edman degradation.

The results obtained were:

a) AVCRTGCKNFL

b) LDCKGGAYKCFRHTKCS

2. After treating the compound with 2-mercaptoethanol, the thiol groups were blocked with iodoacetamide, followed by trypsin digestion. MS/MS analysis was performed to determine the molecular weight of each amino acid in the peptide fragments (all listed from N- to C-terminus):

a) 71.04, 99.07, 160.03, 156.01

b) 160.03, 87.03

c) 101.04, 57.02, 160.03, 128.09

d) 114.04, 147.07, 113.08

e) 137.06, 101.05, 128.09

f) 160.04, 147.07, 156.10

g) 57.02, 57.02, 71.03, 163.06, 128.09

h) 113.08, 115.03, 160.03, 128.09

i) 114.04, 128.06

j) 159.04, 114.04, 115.03, 99.07, 160.03, 156.10

3. After treating the compound with a demethylase and trypsin digestion, the resulting fragments were (Cys2 indicates cystine):

a) Ala, Val, Cys2, Arg, Ser

- b) His, Thr, Lys
- c) Thr, Gly, Cys2, Lys, Phe, Arg
- d) Asn, Phe, Leu
- e) Gly, Gly, Ala, Tyr, Lys
- f) Asn, Gln
- g) Leu, Asp, Cys2, Lys, Arg, Val, Asp, Asn, Met

For different peptide chains within the compound, they are named in order of length from longest to shortest as **Chain A**, **Chain B**, **Chain C**, **Chain D**, and so on. Residues are denoted in the format **PeptideChain-ResidueNumber**, e.g., A3 represents the third residue of **Chain A**, and B9 represents the ninth residue of **Chain B**. Assuming only standard residues and their modifications are considered, and no unnatural amino acids are present, which of the following statements about this peptide compound is correct?

- A.** The compound has three polypeptide chains, with disulfide bond positions at A3-C5, B3-A16, and B7-A10, and a molar ratio of 1:1 in reaction with cyanogen bromide. The compound contains a total of 15 methyl groups.
- B.** The compound consists of three polypeptide chains, with disulfide bond positions at A3-C5, B3-A16, and B7-A10. The molar ratio of reaction with cyanogen bromide is 1:1. The compound contains a total of 13 methyl groups.
- C.** The compound has three polypeptide chains, with disulfide bond positions at A15-C5, B3-A2, and B7-A8. The molar ratio of its reaction with cyanogen bromide is 1:1. The compound contains a total of 15 methyl groups.
- D.** The compound has three polypeptide chains, with disulfide bond positions at A15-C5, B3-A2, and B7-A8, and a molar ratio of 1:1 in the reaction with cyanogen bromide. The compound contains a total of 13 methyl groups.

- E.** The compound has three polypeptide chains, with disulfide bond positions at A3-C5, B3-A16, and B7-A10, and does not react with cyanogen bromide. The compound contains a total of 15 methyl groups.
- F.** The compound has three polypeptide chains, with disulfide bond positions at A3-C5, B3-A16, and B7-A10, and does not react with cyanogen bromide. The compound contains a total of 13 methyl groups.
- G.** The compound has three polypeptide chains, with disulfide bond positions at A15-C5, B3-A2, and B7-A8, and does not react with cyanogen bromide. The compound contains a total of 15 methyl groups.
- H.** The compound consists of 3 polypeptide chains, with disulfide bond positions at A15-C5, B3-A2, and B7-A8, and does not react with cyanogen bromide. The compound contains a total of 13 methyl groups.
- I.** The compound has two polypeptide chains, with disulfide bond positions at B3-A16 and B7-A10, and does not react with cyanogen bromide. The compound contains a total of 10 methyl groups.
- J.** The compound consists of two polypeptide chains, with disulfide bond positions at B3-A2 and B7-A8, and does not react with cyanogen bromide. There are a total of 10 methyl groups in the compound.
- K.** None of the above options is correct

Answer

Correct Answer: A

Detailed Explanation

First, let's examine Experiment 1. Edman degradation yielded the sequences of two peptide chains: AVCRTGCKNFL and LDCKGGAYKCFRHTKCS. Since BME was used to disrupt disulfide bonds in both Experiments 1 and 2, we can temporarily disregard the positions of the disulfide bonds.

We first write down the products of the two peptide chains in Experiment 1 after trypsin digestion and match the sequences with the molecular weights from Experiment 2.

Digest Product	Amino Acid Molecular Weights	Mass Spectrometry Results
AVCR	71.03, 99.07, 160.03, 156.10	2a
TGCK	101.05, 57.02, 160.03, 128.09	2c
NFL	114.04, 147.07, 113.08	2d
LDCK	113.08, 115.03, 160.03, 128.09	2h
GGAYK	57.02, 57.02, 71.04, 163.06, 128.09	2g
CFR	160.03, 147.07, 156.10	2f
HTK	137.06, 101.05, 128.09	2e
CS	160.03, 87.03	2b

2i and 2j could not be matched to any products of the two peptide chains sequenced in Experiment 1, indicating the presence of a third peptide chain composed of 2i and 2j in the compound.

CHECKPOINT

1 PTS

There is a third peptide chain composed of 2i and 2j, meaning the compound includes three peptide chains

The reason this peptide chain was not sequenced may be due to cyclization, N-terminal post-translational modifications, or other factors. Since Edman degradation relies on reacting with the free N-terminal amino group of the protein, it cannot sequence the peptide when the amino group is blocked.

Next, we deduce the sequences of these two fragments. 2i is clearly NQ, while 2j can be partially inferred as NDVCR. However, the first residue with a molecular weight of 159.04 does not correspond to any standard residue. Note that the question states non-natural amino acids should not be considered, so this residue must be a modified form of a standard residue.

CHECKPOINT

1 PTS

The residue with a molecular weight of 159.04 is a modified form of a standard residue

Additionally, as mentioned earlier, Edman degradation could not sequence this peptide chain because its N-terminal amino group was blocked. Therefore, 159.04 corresponds to an amino acid with a blocked N-terminal amino group. Cyclic peptides can be ruled out here because i and j have only one cleavage site (the C-terminal R of j). If it were a cyclic peptide, digestion would yield a single linear peptide, not two. Furthermore, the mass spectrometry results also indicate that this peptide could not be linked to other peptide chains via isopeptide bonds, as the digestion products would otherwise appear together with those sequenced in Experiment 1.

Examining common N-terminal modifications, such as acetylation or methylation, we note that $159.04 = 131.04 + 28$, corresponding to methionine + two methyl groups. Thus, the N-terminal methionine of this peptide is blocked by dimethylation, and the third peptide chain can be written as $(CH_2)_2MNDVCRQN$. The connectivity here is uniquely determined by the digestion results of 2i and 2j.

CHECKPOINT

1 PTS

The N-terminal methionine of this peptide is blocked by dimethylation

CHECKPOINT

1 PTS

The sequence of the third peptide chain is MNDVCRQN, with the amino group of M dimethylated

Now, let's examine Experiment 3. Note that Experiment 3 begins with the use of a demethylase, which aligns with our reasoning in Experiment 2 and follows the progressive logic of Experiments 1–3.

We match the disulfide bond positions identified in Experiment 3 with the peptide chains A, B, and C. Here, LDCKGGAYKCFRHTKCS is chain A, AVCRTGCKNFL is chain B, and MNDVCRQN is chain C. For example, 3a indicates a disulfide bond between Cys3 of chain B and Cys16 of chain A (i.e., AVC-CR), while 3c indicates a disulfide bond between Cys7 of B and Cys10 of A. Additionally, the results of 3f and 3g are consistent with the sequences we deduced in Experiment 2, and one disulfide bond is located between Cys3 of A and Cys5 of C.

CHECKPOINT

1 PTS

The disulfide bond positions are: A3-C5, B3-A16, B7-A10

Cyanogen bromide reacts specifically with the side chain and C-terminus of methionine to form homoserine lactone. Since the compound contains only one methionine in chain C, the reaction stoichiometry is 1:1.

CHECKPOINT

1 PTS

The compound contains 1 methionine, and the reaction stoichiometry with cyanogen bromide is 1:1

Finally, let's count the number of methyl groups. The three peptide chains are: AVCRTGCKNFL, LDCKGGAYKCFRHTKCS, and MNDVCRQN. The residues with methyl groups in their side chains are: A, T, M (1 each), V, L (2 each). Thus, the side chain methyl count is: $1 + 2 + 1 + 2 + 2 + 1 + 1 + 1 + 2 = 13$, plus the two methyl groups at the N-terminus of chain C, totaling 15.

CHECKPOINT

1 PTS

There are 15 methyl groups in total: 13 on side chains and 2 on the N-terminus of the main chain

Option A is correct.